KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI
COLLEGE OF ENGINEERING

AE 451 IRRIGATION AND DRAINAGE

TUTORIAL QUESTIONS

SECTION A

1. Irrigation is……..

2. State 2 reasons *ONLY* why irrigation is important.
a)……
b)……

3. Name the 2 categories of irrigation practice:
   a)
   b)

4. State the concepts mentioned in question 3 in mathematical expressions:
   a)
   b)

5. You are required to do a feasibility study for a proposed surface irrigation system. *List* the data you would require to enable you design the system.

6. A typical irrigation system comprises:
   (a)
   (b)
   (c)

7. What should be the mode of water supply for irrigation when the farm level is *lower* than the source of water?
   (a) gravity
   (b) pumping
   (c) sprinkling
   (d) none of the above

8. What should be the mode of water supply for irrigation when the farm level is *higher* than the source of water?
   (a) gravity
   (b) pumping
   (c) sprinkling
   (d) none of the above

9. List the type of canal structures there are and the functions they perform in the supply of water to the irrigation field.

10. During the construction of an irrigation canal, it is required to carry water across a road to the block of farms located on one side of the road yet at the same time enabling the road to continue to be used. What hydraulic structure will you recommend to achieve this if the level of water in the canal is well below the level of the road to be crossed?

(a) culvert
(b) inverted siphon
(c) Parshall flume
(d) none of the above

11. In question 9, what hydraulic structure will you recommend if the level of the road is below the level of water in the canal?

(a) culvert
(b) inverted siphon
(c) Parshall flume
(d) none of the above

12. Write simple mathematical expressions relating the following:

- Soil total water potential, matric potential, gravitational and osmotic potentials.
- Soil total water potential, hydraulic and osmotic potentials.
- Soil hydraulic potential, matric and gravitational potentials.

Problem: Consider a crop with rooting depth of 1.5 m, grown in a medium textured soil. The field capacity and permanent wilting points (weight basis) are 20 and 9% respectively. The specific gravity of the soil is 1.30, the maximum allowable soil water depletion factor is 0.5, the peak crop consumptive water use is 7.5 mm/day and the irrigation efficiency is 60 %. [***Answer questions 13 – 16***]

13. Calculate the water content of the soil at irrigation (% volume)

(a) 18.9 %
(b) 16.3 %
(c) 14.5 %
(d) none of the above

14. Calculate the net irrigation requirements.

(a) 179.0 mm
(b) 107.3 mm
(c) 71.5 mm
(d) none of the above

15. Calculate the gross irrigation requirements.
    (a) 179.0 mm
    (b) 107.3 mm
    (c) 71.5 mm
    (d) none of the above

16. Calculate the design irrigation interval.
    (a) 24 days
    (b) 18 days
    (c) 14 days
    (d) none of the above

Problem: A sample of soil was taken from a 3 ha farm. The diagram below shows the proportion by mass and volume of air, water and solids in a soil at the time of irrigation. Also, volume of voids, $V_v = V_a + V_w$. Soil particle density is 2.65 Mg/m^3.

	SOIL	
volume of air = V_a	Air	M_a = mass of air
0.00032 m^3 = volume of water = V_w	Water	M_w = mass of water
volume of solids = V_s	Solids	M_s = mass of solids = 1.245 kg
0.001 m^3 = $V_a + V_w + V_s = V$		$M = M_a + M_w + M_s$

Refer to the figure above and answer questions 17 - 23.

17. What is the mass water content of the soil?
    a) 0.320
    b) 0.257
    c) 0.530
    d) None of the above

18. What is the volumetric water content of the soil?
    a) 0.320
    b) 0.257
    c) 0.530
    d) None of the above

19. Calculate the dry bulk density of the soil.
    a) 1.245 Mg/m^3
    b) 1.565 Mg/m^3
    c) 1.408 Mg/m^3
    d) None of the above

20. What is the wet bulk density of the soil?
    a) 1.245 Mg/m^3
    b) 1.565 Mg/m^3
    c) 1.408 Mg/m^3
    d) None of the above

21. Determine the degree of saturation of the soil.
    a) 0.320
    b) 0.530
    c) 0.600
    d) None of the above

22. What is the depth of water present in 1.5 m profile of the soil?
    a) 320 mm
    b) 480 mm
    c) 600 mm
    d) None of the above

23. What is the total volume of water present in 1.5 m depth of the soil at the time of irrigation in the 3 ha farm?
    a) 1,800 m^3
    b) 9,600 m^3
    c) 14,400 m^3
    d) None of the above

24. A field water study for an irrigation farm revealed the following information:

Soil Depth (cm)	Field Capacity (% wt.)	Field water content (% wt)	Soil dry bulk density (g/cm^3)
0 – 10	18.5	7.1	1.52
10 - 20	22.7	8.3	1.35
20 – 30	25.2	10.2	1.20
30 – 40	20.3	8.0	1.39
40 - 50	19.8	7.9	1.43

Compute the amount of water required (mm) to bring the soil water content to field capacity. If the field size is 5 ha and the average field conditions are represented by the data above, what will be the total irrigation water required (m^3) to bring the soil water to field capacity if the field water application efficiency is 90 %? Present your solution in a **TABULAR** form **ONLY**.

25. The gross irrigation required at an irrigation efficiency of 70 % for a sprinkler system was 100 mm. Compute the irrigation interval based on peak crop water demand of 7.0 mm/day.

26. State *two* disadvantages of the *fixed amount, fixed interval method* for irrigation scheduling.

27. In a sprinkler irrigation design, list the full specification of the sprinkler system usually supplied by manufacturers.

a)
b)
c)
d)
e)

28. A sprinkler nozzle discharges at 0.36 l/s. If the lateral **x** sprinkler spacing is 18 m **x** 18 m, what is the sprinkling rate of the sprinkler? Would you consider this nozzle suitable for a soil which has an infiltration rate of 10 mm/h? (*Yes* or *No*).

Problem: You have been contracted by the Tono irrigation project to do cost analysis for the production of irrigation rice. The data generated for such an estimate are as follows:

Investment	Gross Cost (¢ / ha)	Years of use for depreciation
Land	365,000	-
Dam / Reservoir	562,500	25
Pump/ Engine	120,000	20
Pipeline & accessories	375,000	20
Canal system	255,000	20

Assume taxes = 1%; Interest = 2.5%; Insurance = 2% on original costs.

Operation

Land preparation: ¢108,000/ha
Seed: 80kg/ha @ ¢300/kg
Fertilizer: NPK: 10bags/ha @ ¢19,500/bag
Sulphate Ammonia ¢ 6 bags/ha @ 15,000/bag
Other Agrochemical: Herbicide: 81/ha @ ¢10,500/l
Furadan: 20kg/ha @ ¢1,950/kg
Cymbush: 21/ha @ ¢18,000/l
Dithane: 5kg/ha @ ¢7,500/kg
Others: ¢67,000/ha
Water: ¢75,000 per ha
Labour: 250 manhours/ha @ ¢1,500/manhour
Repair and maintenance: Dam/Reservoir 1%; Pump/engine 5%;
Pipe line & accessories 1%; Canal 5% of original costs
Polysacks: 75nos/ha @ ¢600/unit
Other incidentals: ¢18,000/ha

What is the equivalent cost in US$ if US$1.00 = ¢1,100 (1995 rate).

Answer questions 29 - 39

29. What is the annual cost of investment per ha on the land?
    a) ¢ 19,800.00
    b) ¢ 23,587.50
    c) ¢ 34,687.50
    d) ¢ 46,406.25

30. What is the annual cost of investment per ha on the dam/reservoir?
    a) ¢ 19,800.00
    b) ¢ 23,587.50
    c) ¢ 34,687.50
    d) ¢ 46,406.25

31. What is the annual cost of investment per ha on the pipe and accessories?
    a) ¢ 19,800.00
    b) ¢ 23,587.50
    c) ¢ 34,687.50
    d) ¢ 46,406.25

32. What is the annual cost of investment per ha on the canal?
    a) ¢ 19,800.00
    b) ¢ 23,587.50
    c) ¢ 34,687.50
    d) ¢ 46,406.25

33. What is the total annual cost of investment per ha?
    a) ¢ 123,587.50
    b) ¢ 130,687.50
    c) ¢ 134,406.25
    d) ¢ 135,581.25

34. What is the annual cost of operation per ha on seed procurement?
    a) ¢ 19,800.00
    b) ¢ 24,000.00
    c) ¢ 285,000.00
    d) ¢ 375,000.00

35. What is the annual cost of operation per ha on fertilizer procurement?
    a) ¢ 19,800.00
    b) ¢ 24,000.00

c) ¢ 285,000.00
d) ¢ 375,000.00

36. What is the annual cost of operation per ha on labour?
    a) ¢ 19,800.00
    b) ¢ 24,000.00
    c) ¢ 285,000.00
    d) ¢ 375,000.00

37. What is the annual cost of operation per ha on repair and maintenance?
    a) ¢ 28,125.00
    b) ¢ 37,500.00
    c) ¢ 45,000.00
    d) ¢ 75,000.00

38. What is the total annual cost of operation per ha?
    a) ¢ 1,108,000.00
    b) ¢ 1,221,625.00
    c) ¢ 1,134,687.50
    d) ¢ 1,146,406.25

39. What is the total annual irrigation cost on investment and operation per ha?
    a) ¢ 1,118,000.00
    b) ¢ 1,221,625.00
    c) ¢ 1,357,206.25
    d) ¢ 1,446,406.25

40. State the conditions for which a canal is said to have an economic channel section.

41. State the approximate relationship between the bed width (*b*) and the depth of flow (*d*) for an economic trapezoidal canal section.

Problem: Design a lined rectangular open drain to carry 0.93 m^3/s for the following conditions: Bed width = 1.4 m; Canal bed slope = 1:500; Manning's roughness coefficient = 0.015. Assume a freeboard of 20%. ***Answer the following questions***.

42. What is the depth of flow?
    a) 0.25 m
    b) 0.30 m
    c) 0.35 m
    d) none of the above

43. What is the velocity of flow?
    a) 0.644 m/s
    b) 0.544 m/s
    c) 0.344 m/s
    d) none of the above

44. 25. State whether the following statements are ***True*** or ***False*** concerning the benefits of land drainage:

    a) Good drainage leads to a limited choice of crops for the land.
    b) A well-drained soil warms up more quickly.
    c) Lower water table results in an increased volume of soil available for plant growth.
    d) The risk of causing damage to the soil structure by livestock walking on wet soil is considerably reduced when the soil is well-drained.
    e) Well-drained land increases the incidence of animal diseases that thrive in wet conditions.
    f) Good drainage reduces the number of occasions when cultivation is held up waiting for the soil to dry up.
    g) Good drainage allows early and good quality harvest.
    h) Land cultivation in poorly-drained soils cause less damage to soil structure.

45. The purpose of *subsurface* drainage is to:

    a) control excess water on the soil surface.
    b) control excess water in the soil and lower the water table
    c) control soil and plant diseases.
    d) None of the above

46. Which of the following is *not* a surface drainage system?

    a) Interceptor drains
    b) Diversion drains
    c) Mole drains
    d) Randomised drains

47. In the construction of open drains, name 4 site conditions that affect the type of equipment to be used.
    a)
    b)
    c)
    d)

Problem: A subsurface drainage system is to be installed in an irrigated area to facilitate leaching. Determine the required drain spacing according to the *Houghoudt* Method for steady state conditions for the following fields:

Parameters	Field 1	Field 2
k_1	0.38m/day	0.50m/day
k_2	1.48 m/day	1.00m/day
m	2.40m/day	3.0m
h	0.50m	0.60m
q	7mm/day	5mm/day
r_0	0.10m	0.1m

Answer questions 48 and 49.

48. Determine the drain spacing for *field 1*.
    a) 38.2 m
    b) 40.6 m
    c) 48.2 m
    d) none of the above

49. Determine the drain spacing for *field 2.*
    a) 38.2 m
    b) 40.6 m
    c) 48.2 m
    d) none of the above

50. State 2 parameters (and their units) that are used to describe salt-affected lands for the purposes of their classification.
    a)
    b)

51. What is the condition for which a soil is said to be saline-sodic?

Problem: Classify the following soils for the purpose of their reclamation, if necessary, using the data obtained from their analysis:

Soil A

ECe = 4.85 mmhos/cm Ca^{2+} = 1.41 mmol/l

Mg^{2+}= 1.01 mmol/l Na^{2+} = 13.5 mmol/l

Soil B

ECe = 3.10 Ca^{2+} = 1.42

Mg^{2+} = 1.21 Na^{2+} = 21.50

Answer questions 35 and 36.

52. Soil A can be classified as:
    a) saline
    b) sodic
    c) saline-sodic
    d) normal

53. Soil B can be classified as:
    a) saline
    b) sodic
    c) saline-sodic
    d) normal

54. State *2* harmful effects of *sodicity* on crop production.

55. Name the parameters that are used to describe salt-affected lands for the purposes of their classification.

56. *SAR* is a function of *Na*, *Ca* and *Mn* ions. (***True*** or ***False***?). At what level of *SAR* (***value and units quoted***) is a soil considered to have a sodicity problem? In one word, state how you will control the harmful effects of sodicity in the soil.

57. One harmful effect of soil sodicity problems is that sodium ions tend to seal up the soil pores thereby reducing the permeability of the soil. (***True*** or ***False***?)

58. A soil is said to be *sodic* if it fulfils which condition of soil extract concentration?

    a) > 4 mmhos/cm; > 13 $(mmol/l)^{1/2}$
    b) > 4 mmhos/cm; < 13 $(mmol/l)^{1/2}$
    c) < 4 mmhos/cm; > 13 $(mmol/l)^{1/2}$
    d) none of the above

59. A soil is said to be *normal* if it fulfils which condition of soil extract concentration?

    a) > 4 mmhos/cm; > 13 $(mmol/l)^{1/2}$
    b) > 4 mmhos/cm; < 13 $(mmol/l)^{1/2}$
    c) < 4 mmhos/cm; > 13 $(mmol/l)^{1/2}$
    d) none of the above

Problem: Given below is the data for hydraulic conductivity of a multi-layered soil in a field, with parallel tube drains of radius 0.1 m installed at 1.0 m below ground level. The average rooting depth of crop to be grown on the field is about 0.75 m.

Soil depth (m)	Hydraulic conductivity (m/day)
0-0.4	0.5

0.4-0.7	0.3
0.7-1.0	0.6
1.0-1.3	0.6
1.3-2.0	1.2
≥2.0	Impermeable layer

Answer questions 60 to 62.

60. What is the hydraulic conductivity above the drain?
    a) 0.47 m/day
    b) 0.60 m/day
    c) 0.64 m/day
    d) 1.18 m/day

61. What is the hydraulic conductivity below the drain?
    a) 0.47 m/day
    b) 0.60 m/day
    c) 0.64 m/day
    d) 1.18 m/day

62. What drain spacing is required to maintain the water table below the root zone if drainage coefficient is 5 mm/day?
    a) 45 m
    b) 36 m
    c) 27 m
    d) 18 m

63. To minimize serious erosion and to stabilize the drain bottom and side walls, vegetation should be established on these areas as soon as construction is completed (*True or False*?).

64. Maintenance of surface drains is a continuing requirement if the drain is expected to continue to perform as designed. The types of maintenance procedures are:

a) corrective maintenance
b) preventive maintenance
c) preventive and corrective maintenance
d) None of the above

SECTION B

65. Explain, *briefly in a logical sequence*, what you would do in the design of a dam and a field layout for surface irrigation system. Your answer must include the things you would do from the detailed feasibility study stage through to the preparation of a complete design portfolio.

66. Calculate the *maximum 24-hr* stream flow (m^3/s) for a proposed 100 ha sugarcane farm using the following data. The sugarcane is planted in September and takes 13 months to mature for harvesting. Assume a furrow irrigation efficiency of 70% and system flexibility factor of 1.1. The calculation must be clearly presented in a table on monthly basis.

Growth period (month)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**
Crop coefficient	**0.3**	**0.5**	**0.7**	**0.9**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**1.0**	**0.9**	**0.7**

Month	**J**	**F**	**M**	**A**	**M**	**J**	**J**	**A**	**S**	**O**	**N**	**D**
ETo (mm)	**261**	**235**	**185**	**169**	**210**	**185**	**210**	**187**	**177**	**210**	**220**	**220**
Pe (mm)						**30**	**136**	**162**	**50**			

67. A distribution system is to be designed for a sprinkler system on which 5 laterals will be operated simultaneously. Each lateral is to have 30 nozzles with 0.360 l/s discharge per nozzle. The required pressure on the mainline at the inlet to each lateral is 393 kPa. The elevation of each point on the line and distance between points are given in the following table:

Point	Elevation (m)	Distance between points (m)
Pump	48.41	
		36.58
A	47.55	
		18.29
B	48.46	
		18.29
C	48.77	
		18.29
D	47.85	
		18.29
E	47.24	

The main is made of 20.32 cm inside diameter aluminium pipe with a Hazen-Williams C factor of 144. The friction loss and elevation head from the water surface to the centerline of the pump is +0.333m. Laterals are to operate on points A through E simultaneously. Determine the critical pressure point in the mainline and the total dynamic head requirement of the pump.

68. (a) Explain how you will determine the resultant hydraulic conductivity above and below the drain in multi-layered soil (each layer having different hydraulic conductivity) for use in the Hooghoudt steady-state drainage equation.

(b) Below is the drain for hydraulic conductivity of multi-layered soil with tube drain of radius 0.1m installed at 1.0m below ground level. The mid-drain water table height is above the drain is 0.25m. What drain spacing (m) is required to control the water table if drainage coefficient is 5mm/day?

Soil depth (m)	Hydraulic conductivity (m/day)
0-0.4	0.5
0.4-0.7	0.3
0.7-1.0	0.6
1.0-1.3	0.6
1.3-2.0	1.2
2.0	Impermeable layer

69. The data below was collected on a pilot scheme using the drain outflow method for the determination of hydraulic conductivity.

q(mm/day)	H(m)
3.31	0.56
2.99	0.59
2.38	0.65
2.03	0.71
0.98	0.82
1.9	0.70
0.78	0.85
4.00	0.50
1.66	0.73
2.13	0.70
0.85	0.84
0.48	0.90

If drains running half full, with wetted perimeter of 0.38m and drain spacing of 25m were used, determine by appropriate method hydraulic conductivities above and below the drain. The drain was placed at the interface of the topsoil and subsoil. Assume a drain depth = 1.2m from the surface and depth from drain to impermeable layer below = 2.8m. Note that q is the specific flow rate and H, the depth to water table at mid-drain position.

70. Chemical analysis for irrigation water to be used for citrus production is as follows:

SAR_{iw} = 2.59 $(mmol/l)^{1/2}$
$[Ca^{2+}]$ = 6.58 meq/l
$[Mg^{2+}]$ = 8.48 meq/l
$[HCO_3^-]$ = 4.39 meq/l
∑[cations] = 22.3 meq/l

Compute the required leaching requirement for sodium control, given further that $ESP_{iw} = 0.05$; $pK_2 + pK_s = 2.36$; $p([Ca^{2+}] + [Mg^{2+}])/2 = 2.12$; $p([HCO_3^-] + [CO_3^{2-}]) = 2.36$. Also compute the required time of leaching to manage the sodium salts if the net irrigation requirement is 75 mm and the soil average infiltration rate is 7.58 mm/h.